1.3 Uniaxial compression test (UAC)

Keywords: plain concrete, compressive softening, fracture localization, prescribed-displacement boundary condition, Newton-Raphson method

Input files: UAC\UAC050.cc2 (50 mm specimen)
UAC\ UAC100.cc2 (100 mm specimen)
UAC\ UAC200.cc2 (200 mm specimen)

1.3.1 Introduction

This example shows how ATENA treats compressive softening and the localization of fracture, which is associated with it.

Van Mier [1] conducted uniaxial compression tests on concrete prisms with identical cross-sections but various heights 50, 100 and 200 mm. The experiments indicated that all the specimens failed by formation of inclined planes of localized fracture. While in the 200 mm high prism, there was a large continuous failure plane, the 100 and 50 mm prisms shoved several intersecting planes. However, the experiments revealed that the amount of energy required for fracturing a specimen was independent of its height (in a similar way as under uniaxial tension). Furthermore, the experiments showed that, if displacement corresponding to elastic deformation was subtracted from the post-peak part of each load-displacement curve, the resulting curves were almost identical for all three specimens.

ATENA allows treating these phenomena by directly inputing the compression softening stress-displacement relationship as a material property, in a similar manner as it is done for tension. Use of the concept is demonstrated by reproducing the van Mier's experiments in this example.

1.3.2 Comments on FE model preparation

1.3.2.1 Topology

We analyze three plain concrete specimens with heights 50, 100 and 200 mm and cross-section of 100x100 mm. Plane stress is assumed. An uniform mesh is used. The element size is 25 mm and is the same for all the specimens. To facilitate fracture localization, a band of material, which is slightly more brittle under compression, is inserted in each specimen (Figure 1-5). The height of this band may be arbitrary but should correspond to the height of a single element.

1.3.2.2 Materials

Concrete is modeled by the 'Nonlinear Cementitious' material model. Relevant material properties used in the analysis are listed in Table 1.3-1.

1.3.2.3 Loads and supports and Run

It is assumed that friction between loading plates and a concrete specimen is

perfectly removed. The bottom line of a specimen is fixed in vertical direction while the top line is loaded by prescribed uniform displacement. In addition, the bottom left corner is fixed horizontally. Since we control displacement, we may employ the Newton-Raphson solution method to trace the response up to the post-peak regime. Relevant options and parameters are summarized in Table 1.3-2.

1.3.3 Results

In all three specimens we can observe almost uniform deformation taking place up to the peak load. After the peak, the axial deformation localizes into the horizontal band of more brittle elements. The deformation of this band further induces lateral tension to the adjacent elements, causing their splitting in the direction parallel with the applied load. The failure pattern is documented in Figure 1-9 for the 200 mm specimen. This cracking behavior differs from that reported by van Mier [1], who observed formation of inclined failure planes. Nevertheless, graphs in Figure 1-6, Figure 1-7, and Figure 1-8 show, that the present approach successfully captures the overall stress-strain behavior for specimens of various heights.

1.3.4 References

[1] Van Mier, J.G.M., Strain-softening of concrete under multiaxial loading conditions, Ph.D. Dissertation, Technical University of Eindhoven, 1984

Table 1.3-1 Material properties

Material type		Nonlinear cementitious	
Elastic modulus	E_c	38	GPa
Poisson's ratio	ν	0.2	-
Compressive strength	f_c	43.9	MPa
Tensile strength	f_t	2.987	MPa
Fracture energy	G_f	74.67	N/m
Plastic strain at strength f_c	ε_{cp}	9.4×10^{-4}	
Onset of non-lin. behavior in compression	f_{c0}	30	MPa
Critical compressive disp. (loc. band)	w_d	5×10^{-4}	m
Critical compressive disp. (rest of spec.)	w_d	6×10^{-4}	m

Table 1.3-2 Solution parameters

Solution method	Newton-Raphson	
Stiffness/update	Tangent/each iteration	
Number of iterations	50	
Error tolerance	0.001	
Line search	on, with iterations	

Table 1.3-3 Finite element mesh

Finite element type	Quadrilateral, CCIsoQuad	
Element shape smoothing	on	
Geometrical nonlinearity	on	
Optimization	Sloan	

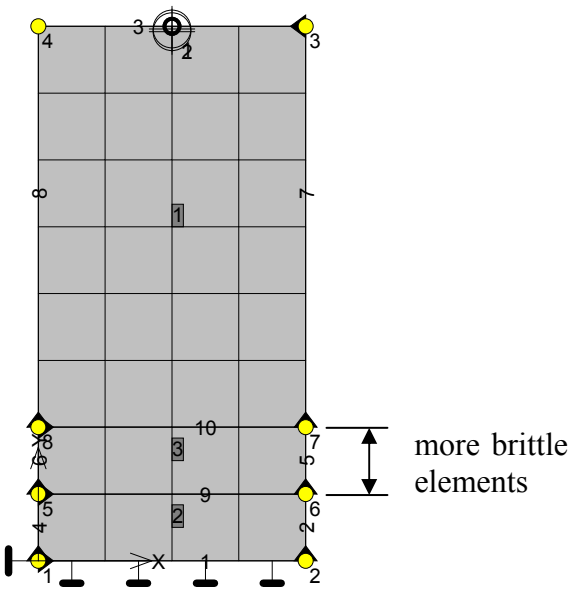


Figure 1-5: Finite element model of uniaxial compression specimen (200 mm high)

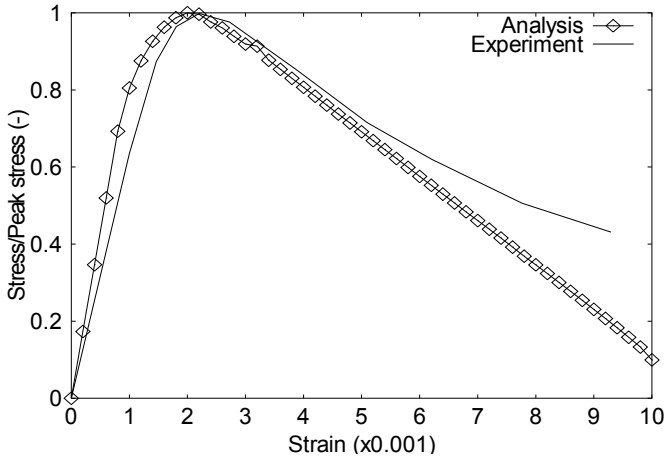


Figure 1-6: Overall stress-strain curves of 50 mm specimen

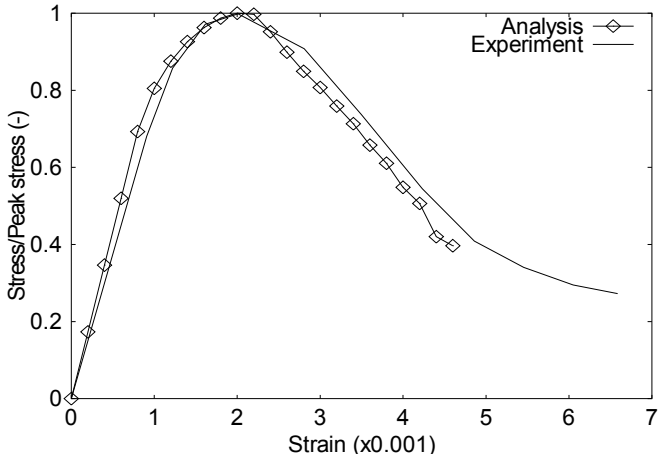


Figure 1-7: Overall stress-strain curves of 100 mm specimen

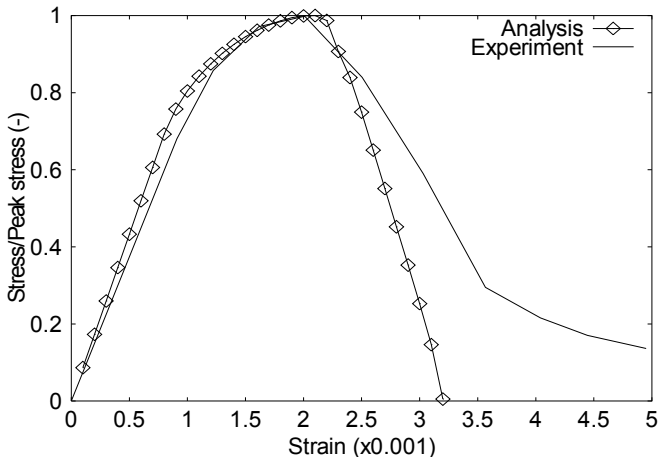


Figure 1-8: Overall stress-strain curves of 200 mm specimen

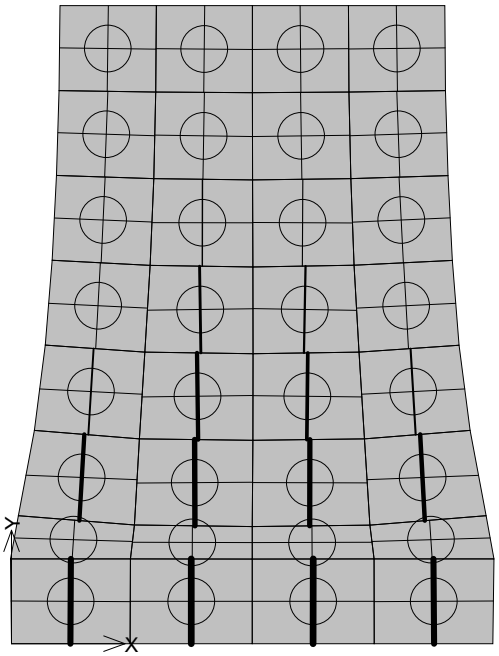


Figure 1-9: Computed crack pattern and deformed shape of the 200 mm specimen after the peak